



CHAPTER

The Dawn of a Maturing Industry

**ELECTRONIC COMMERCE
From Vision to Fulfillment**

Third Edition

Elias M. Awad

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The focus of this chapter is on several learning objectives

- **Conceptual understanding of e-commerce, e-business and e-strategy**
- **Advantages and Issues in E-Commerce**
- **Value-chain and supply-chain management and how they relate to e-commerce and e-business**
- **Business models of the e-environment**
- **A trend toward integrating e-commerce**



What Is E-commerce?

- **Communications Perspective** : the ability to deliver products, services, information, or payments via networks.
- **Interface View** : E-commerce means information and transaction exchanges between:
 - Business-to-Business (B2B)
 - Business-to-Consumer (B2C)
 - Consumer-to-Consumer (C2C)
 - Business-to-Government (B2G)

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What Is E-commerce?

- **Business Process** : E-commerce means activities that support commerce electronically by networked connections.
- **Online Perspective** : E-commerce is an electronic environment that allows sellers to buy and sell products, services, and information on the Internet.
- **A Structure** : E-commerce deals with various media: data, text, Web pages, Internet telephony, and Internet desktop video.
- **A Market** : E-commerce is a worldwide networked Market.

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E-business

- Sometimes used interchangeably to refer to e-commerce
- E-business covers online transactions, but also extends to all Internet based interactions with business partners, suppliers and customers such as:
 - selling direct to consumers,
 - manufacturers and suppliers;
 - monitoring and exchanging information;
 - auctioning surplus inventory;
 - collaborative product design.
- aimed at improving or transforming business processes and efficiency.

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E-commerce Is Not E-business

- *E-business* - the conduct of business on the Internet, in supply-chain planning, tracking, fulfillment, invoicement, and payment.
 - Includes buying and selling as well as servicing customers and collaborating with business partners
 - Electronic information is used to boost performance and create value by forming new relationships between and among businesses and customers.
 - One example of e-business is SAP (www.sap.com)



E-commerce Versus E-business

- **E-commerce**

- Selling goods and services on the retail level with anyone, anywhere, via the Internet
- Greater efficiency and effective exchange of goods and services
- Exchange based upon *transaction*
 - A block of information exchange between the merchant and its customers via the corporate Web site
 - www.amazon.com

- **E-business**

- Connecting critical business systems and constituencies directly via the Internet

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The Drivers for E-commerce

- **Digital Convergence** - digital devices communicating with one another
- **Anytime, anywhere, anyone** - e-commerce is available to anyone (24/7)
- **Changes in Organizations** - today's businesses empower frontline workers to do the kind of work once performed by junior management
- **Increasing pressure on operating costs and profit margins** - global competition and the proliferation of products and services worldwide have added pressure on operating costs and profit margins
- **Demand for customized products and services** - mass customization puts pressure on firms to handle customized requests on a mass-market scale

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Myths about E-commerce

- Setting up a Web site is easy.
- E-commerce means no more mass marketing.
- E-commerce means a new economy.
- E-commerce is revolutionary.
- E-commerce is a commercial fad that crashed in 2000.
- All products can be sold online using identical business models.
- Build it and they will come.
- The middleman is out.



Advantages through E-commerce

- **Lower cost to the E-merchant**
 - Cost effective
 - Reduces logistics problem
 - Give small company equal exposure as the big companies
 - Lower transaction processing charge
- **Economy**
 - No rental of physical storage space, insurance or infrastructural investment
 - Only need a good idea, unique product and well designed storefront
- **Higher margins**
 - Businesses gain more control and flexibility when manual transaction is done electronically
 - Low transaction processing charge



Advantages through E-commerce

- **Better customer service**
 - Web based customer service make user happy
 - Let user serve him/herself save time and money
- **Quick comparison shopping**
 - E commerce allows comparison shopping using automated shopping assistant program called *shopbot*
 - shopbots finds best deals from web storefronts
- **Productivity gains**
 - Weaving the web tech. throughout the organization means improved productivity
- **Teamwork**
 - Supports user collaboration in exchanging information and work on solution
 - More interaction mean better overall result.
 - improved communication with its suppliers, vendors and business partners

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Advantages through E-commerce

- **Information sharing, convenience, and control**
 - Improves information sharing between merchants and customer and promote quick, just-in-time deliveries
 - Open 24x7, no traffic jam, do not have to carry shopping bags
 - Business can give more control to his customer managing his account
- **Customization**
 - Digital products are highly customizable
 - Easy to recognize, reverse or edit
 - With the information about the customer products can be customized and matched to individual needs



Issues and Constraints of E-commerce

- The cost factor
- Security
- System and data integrity
- System scalability
- E-commerce is not free
- Fulfillment and customer relations problems
- Products people resist buying online
- Cultural, language, and trust issues
- Corporate vulnerability
- Lack of a blueprint for handling E-commerce
- High risk of Internet start-up

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Benefits and Limitations of the Internet

- The Internet is the enabler of the e-commerce
- Marketing and selling products and services
- Doing business fast
- Gathering opinions and trying out new ideas
- Leveling the playing field
- Promoting a paper-free environment
- Providing superior customer service and support resources
- Efficiency and unequaled cost-effectiveness
- Supporting managerial functions, spreading ideas, ease of technical support
- Triggering new business
- Providing Web services

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Limitations of the Internet

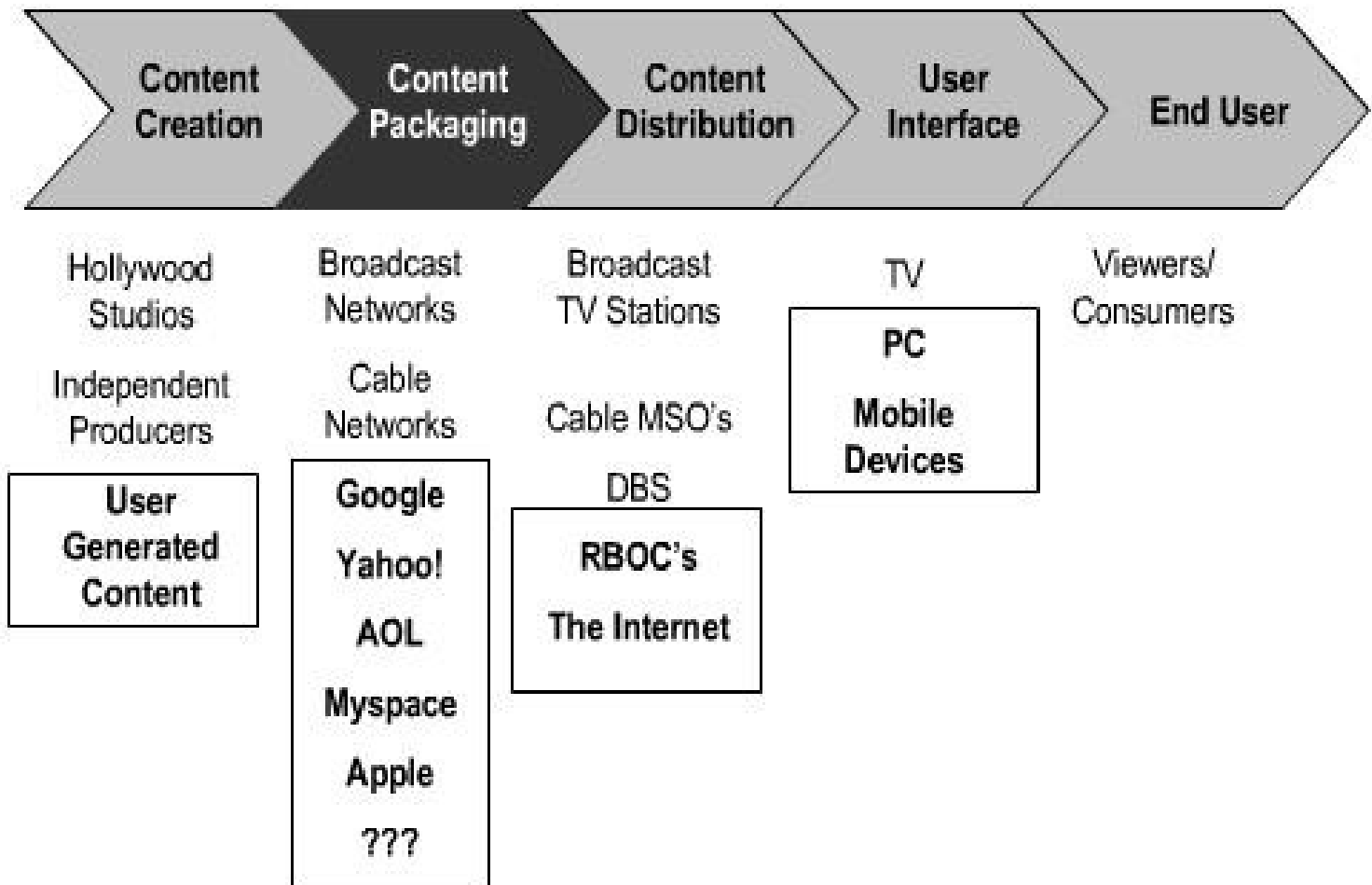
- **Security and privacy**
- **Fakes and forgeries**
- **Cyber terrorism**
- **Problems and stress**
- **Abuses in the workplace**

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Value Chain in E-commerce

- **Value Chain:** A way of organizing the activities of a business so that each activity adds value (value-added activity) or productivity to the total operation of the business.
- A strategic tool for identifying how the critical components of a business tie together to deliver value for the business across the value-chain process.
- Makes B2C interface operational

Value Chain



Value Chain Activities





Value Chain Primary Activities

- **Inbound logistics**

- Procurement Activities: vendor selection, comparative shopping, negotiating supply contract
- Represents supply side of the business

- **Operations**

- Conversion of raw material into finished product
- Machining, packaging, assembly, equipment maintenance, testing etc.
- Provides added value for the marketing function.
- Operational activities are the point in the value chain where value is added



Value Chain Primary Activities

- **Outbound logistics**
 - Storing, distributing and shipping the final product.
 - Involves warehousing, material handling, shipping and timely delivery to the retailer
 - Output of this activity is tied directly to marketing and sales
- **Marketing and sales**
 - Deals with ultimate customers
 - Includes advertising, product promotion, sales management, identifying customer base and distribution channel
 - Output of this activity triggers
 - increased production,
 - more advertising



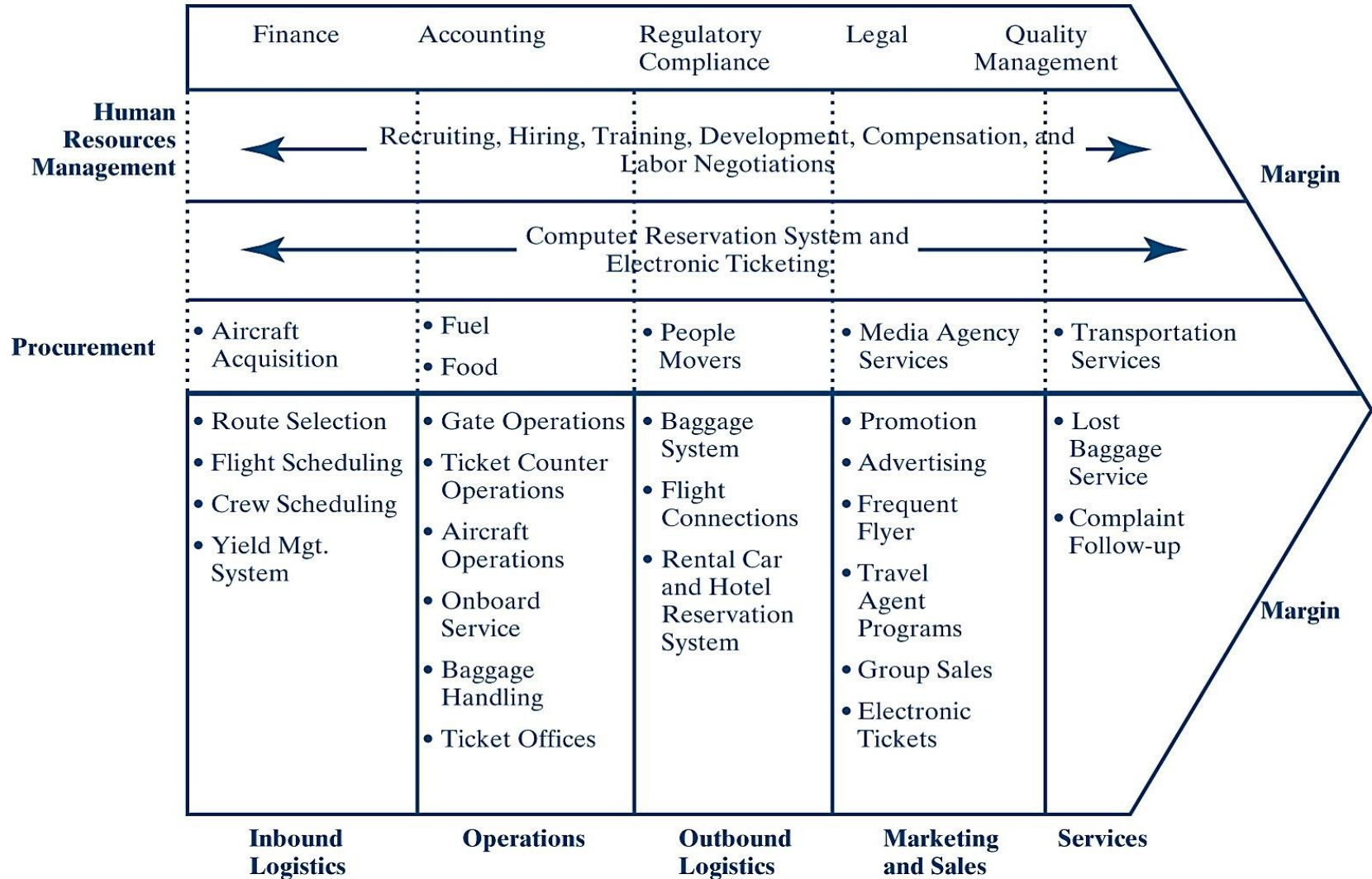
Value Chain Primary Activities

- **Service**

- Focus is on after sales service
- Includes testing, maintenance, repair, warranty works and replacement parts
- Output of this activity means
 - satisfied customer,
 - improved image of the product and the business and
 - potential for increased production

Value Chain for American Airlines

The Value Chain for American Airlines





Value Chain Support Activities

- **Corporate infrastructure**
 - General management, accounting, finance, planning and legal service, quality management
- **Human resources**
 - Activity to matching right people to the job
 - Involves recruitment, retention, career path development, compensation, training and development etc.
- **Technology development**
 - Improves the products and business processes
 - Output: product quality, integrity and reliability
- **Procurement**
 - Purchasing, ensuring availability of quality raw materials

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Analyzing Value Chain Activities

- What type of activity is being performed? Does it add value? Does it ensure the quality of other activities?
- How does the activity add value to the customer?
- Could the same activity be reconfigured or performed in a different way?
- What inputs are used? Is the expected output being produced?
- Is the activity vital? Could it be outsourced, deleted completely, or combined with another activity?
- How does information flow into and out of the activity?
- Is the activity a source of competitive advantage?
- Does the activity fit the overall goals of the organization?

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E-commerce Value Chain

- The E-commerce Value Chain means identifying:
 - The competitive forces within the company's e-commerce environment
 - The business model it will use
 - Identifying the value activities that help the e-commerce value chain do its homework
- E-commerce views information technology as part of a company's value chain

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Roles for E-commerce

- Reducing costs
- Improving product quality and integrity
- Promoting a loyal customer base
- Creating a quick and efficient way of selling products and services
- Incorporate information technology and telecommunications to improve overall productivity
- Web sites are used to provide and collect information between the e-merchant and the customer
 - Competitive pricing information
 - Invoicing facilitates online payment flows
 - Customized products or orders can be shipped and delivered by independent shippers directly to the customer

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E-commerce Application

- **Business-to-Consumer (Internet)**
- **Business-to-Business (Internet and Extranet)**
 - **Supply-Chain Management**
- **Business-within-Business (Intranet)**
- **Business-to-Government (B2G)**

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Business-to-Consumer (Internet)

- The consumer's use of a merchant's Web storefront or Web site
- Modeled on the traditional shopping experience
- Shopping cart is used to hold goods until the customer is ready to check out
 - Online order form supported by the appropriate software
- **Checkout** is order and payment processing

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Business-to-Business (Internet and Extranet)

- **Business-to-Business** e-commerce is industrial marketing. among the processes it handles are fulfillment and procurement
- Companies can conveniently and quickly check their suppliers' inventories or make instant purchases
- B2B often use an extranet: a shared intranet vendors, contractors, suppliers, and key customers

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Supply-Chain Management

- Integrating the networking and communication infrastructure between businesses and suppliers
- Having the right product in the right place, at the right time, at the right price, and in the right condition
- Delivery of customer and economic value through integrated management of the flow of physical goods and related information
- Designed to improve organizational processes by optimizing the flow of goods, information, and services between buyers and suppliers in the value chain

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Supply-Chain Management – cont'd

- **SCM is:**
 - ***Collaboration*** among business partners
 - ***Coordination*** of logistics for timely delivery of goods or products
 - ***Cooperation*** among businesses and suppliers to make sure orders and inquiries are filled correctly
 - ***Connectivity*** through networking infrastructure to ensure speed and good response time at all times

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Business-within-Business (Intranet)

- Strictly a “within company” type of information exchange
 - Restricted to internal employees and customers
 - Firewalls to keep out non employees
 - E-mail replaces paper for the communication of messages, order acknowledgement and approvals, and other forms of correspondence
 - The intranet becomes a facilitator for the exchange of information and services among the departments or divisions of a company
 - Different departments with different PCs or local area networks can interact on an intranet

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Benefits of an Intranet

- **Low development and maintenance costs**
- **Environmentally friendly because it is company-specific**
- **Availability and sharing of information**
- **Timely, current information**
- **Quick and easy dissemination of information**

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Business-to-Government (B2G)

- Online procurement
- Ensuring transparency, reducing cost
- The government market is similar to B2B.
- difficulties
 - Changing the status quo in government is not so easy
 - Changes to employee tasks and job restructuring often create resistance
 - Tax savings potential is not easily recognized
 - Committing to technology means constant need for upgrades and additional costs

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Chapter Summary

- **Electronic commerce (EC) is the ability to deliver products, services, information, or payments via networks such as the Internet and the World Wide Web.**
- **Electronic business connects critical business systems directly to key constituents**
- **The rise of specialized Web sites and Web logs (blogs) generated opportunities to read and write on a vast array of topics**
- **Several drivers promote EC: digital convergence**
- **Advantages of EC**

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Chapter Summary (continued)

- **Limitations of EC**
- **A value chain is a way of organizing the activities of a business so that each activity provides added value or productivity to the total operation of the business.**
- **The transaction life cycle includes three major e-commerce applications: Business-to-Consumer (B2C), Business-to-Business (B2B), and Business-within-Business.**
- **An intranet wires the company for information exchange.**
- **Success in the E-commerce field depends on attracting and keeping qualified technical people and managerial talent.**
- **There are several types of specialized Web sites on the Internet. Each site is based on a business model as a way of doing business to sustain a business - generated revenue.**